

William Gao

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EDUCATION

Carnegie Mellon University, College of Engineering

Pittsburgh, PA

B.S. in Mechanical Engineering with Minor in Robotics: GPA: 3.70/4.0

Aug 2024 - May 2028

Relevant coursework: Intro to Electrical and Computer Engineering, Fund. of Programming and Comp. Sci., Thermodynamics, Mechanics I: 2D Design, Mechanics II: 3D Design, Differential Equations, Fluid Mechanics, Principles of Imperative Computation

In-progress coursework: Heat Transfer, Dynamics, Numerical Methods, Mechanical Design: Methods and Applications

CAD/Design Skills: Solidworks, 3D Printing, Laser Cutting, Ansys Discovery, Bambu Studio, FEA

Programming/Hardware Skills: Python, Java, Arduino board and IDE, Matlab, C

PLC/Automation: Do-More Designer, Ladder Logic

WORK EXPERIENCE

Sepro America

Pittsburgh, PA

Engineering Intern

May 2026 - Aug 2026

- Engineered PLC-based tote inspection sequence (RFID, barcode, OCR verification across dual sides) for a 16-station production scale-out, accounting for ~15 distinct fail conditions and reducing cycle time from 40-60s to under 25s
- Designed and validated a new FANUC robot riser attachment for a new base configuration, achieving a minimum factor of safety of 4-5 via structural FEA in Hilti PROFIS Engineering
- Authored complete cobot palletizing algorithm accounting for X/Y/Z moment shifts across single- and dual-tote two-part gripper configurations, dynamic stack-height placement, and dual 0°/90° tote orientations
- Fabricated and wired 20+ industrial cables via crimping and soldering, and wired the robot control system by following electrical drawings, applying a fundamental understanding of relay, transformer, and AC/DC power concepts, saving the company \$10,000+ in outsourced labor

Carnegie Mellon University

Pittsburgh, PA

Teaching Assistant - Physics I for Engineering Students

Aug 2025 - Dec 2025

- Supported 20+ students by offering 1:1 support and grading homework and exams
- Taught students concepts in kinematics, thermodynamics, rotational dynamics, work, power, and momentum collisions, leading to improved understanding and performance on exams

PROJECTS

Topology-Optimized Structural Bracket Design – Carnegie Mellon University

Fall 2025

- Iteratively designed and weight-optimized an 8-inch structural bracket in SolidWorks under fixed and roller support constraints, focusing on aggressive material reduction while maintaining load capacity
- Validated designs using ANSYS Discovery FEA (von Mises stress, FoS 1–2) and physical laser-cut acrylic testing, converging on a sub-10-gram bracket that successfully supported a 40-lb load

2D Game Co-Development (Galaxy Blitz) - Carnegie Mellon University

Spring 2025

- Developed an interactive 2D platform game using Python, incorporating sprite animations, game physics, player/environment interactions, resolution mapping, and fundamental coding logic

Collaborative Mechanical Design and Fabrication - Carnegie Mellon University

Fall 2024

- Collaborated with a 4-person team to design and fabricate a fully functional marble dispensing mechanism using SolidWorks, laser-cut wood components, 3D-printed components, and a DC motor drive system
- Performed stress and motion simulations in SolidWorks/ANSYS Discovery to validate mechanical stability and optimize part geometry, ensuring smooth and consistent dispensing operation

LEADERSHIP & ACTIVITIES

Varsity Swim Team & Endurance Athletics - Carnegie Mellon University

Aug 2024 - Present

- Ranked as a top 7 fastest swimmer in program history across 4 events; received CSCAA Scholar All-America award
- Trained and successfully completed an Ironman 70.3 Triathlon (Jun 2026)

Board of Directors, Student Athlete Representative – Allegheny Mountain Swimming

Aug 2022 - Aug 2024

- Organized an interactive fundraiser with ~100 participants, raising \$1,000+ to support athlete travel for training camps
- Coordinated and executed a ~50-person social event within a \$2,500 budget to foster cross-team athlete bonding

Competitive Swimmer at National Level - PEAQ Swim Club / North Allegheny High School Swim Team

Jun 2018 - Aug 2024

- 2X USA Swimming Scholastic All-American; 6X NISCA All-America; 6X WPIAL regional champion; 6X PIAA state finalist